Project: P2P Payment Ledger (Simplified UPI/Wallet)

Background

Build a peer-to-peer digital wallet application where users can register, load money, transfer money to other users, and track their transactions. All transfers follow an ACH-style delayed settlement model — money leaves the sender immediately but is only credited to the receiver after a 3-business-day holding period. The system should detect basic fraud patterns, notify users of important events, run background settlement and housekeeping jobs, and provide an admin view for monitoring.

Timeline: 1 week

Tech Stack: Spring Boot 3.x, Spring Data JPA, Spring Security (JWT), Spring Scheduler, PostgreSQL, React

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Functional Requirements

1. User Registration & Authentication

- A user registers with name, email, phone number, password, and a 4-digit transaction PIN.
- On successful registration, the system must automatically create a wallet for the user with ₹0 balance.
- Users log in using email and password. The system issues a JWT token on successful login.

- The system must track failed login attempts. After 5 consecutive failed attempts, the account must be locked for 30 minutes.
- There are two roles — USER and ADMIN. Regular users can only access their own data. Admins have access to the admin panel.
- The transaction PIN is separate from the login password and is required for every money transfer.

2. Wallet

- Every user has exactly one wallet, created automatically during registration with ACTIVE status.
- Users can add money to their wallet. This is a simulated operation — no real bank integration. The system simply credits the amount to the wallet.
- The wallet must display the **available balance** — money the user can spend or transfer right now.
- The wallet must also display today's total spending.
- The wallet has three statuses:
  - **Active** — wallet is fully functional. User can send and receive money.
  - **Inactive** — user has self-deactivated their wallet. All outgoing transactions are blocked. Incoming settlements (money being sent to this user) are declined — the transaction is marked DECLINED, the sender's available balance is refunded, and the sender is notified. The user can reactivate their wallet back to Active at any time.
  - **Frozen** — only an admin can freeze and unfreeze a wallet. All outgoing transactions are blocked. Incoming settlements are declined — the transaction is marked DECLINED, the sender is refunded and notified. The user cannot change this status themselves.
- When a user tries to send money from an Inactive or Frozen wallet, the transaction must be rejected with a clear error message indicating the wallet status.
- When a pending settlement is due for a receiver whose wallet is Inactive or Frozen, the settlement job must decline the transaction, refund the sender, and notify the sender that the transfer was declined.

3. Send Money (ACH — Delayed Settlement)

- A user can search for another registered user by phone number or email.
- To send money, the user selects a recipient, enters an amount, adds an optional remark, and confirms with their 4-digit transaction PIN.
- When a transfer is initiated:
  - The sender's available balance is reduced immediately (money leaves the sender's wallet right away).
  - A DEBIT ledger entry is created for the sender at this point.
  - The receiver's wallet is **not** credited immediately. The transfer amount is held internally by the system until settlement.
  - The transaction status is set to PENDING.
  - The transaction records a `settlement_due_date` which is 3 days from the initiation date.
- Only the sender can see the pending transfer in their transaction history with status "Pending" and the expected settlement date. The receiver is not aware of the transfer until it is settled and credited to their wallet.
- A background settlement job (see Section 7) processes pending transfers and credits the receiver's available balance once the settlement date is reached.
- On settlement:
  - The receiver's available balance is increased by the amount.
  - The CREDIT ledger entry for the receiver is created at this point.
  - The transaction status changes from PENDING to SUCCESS.
  - Both sender and receiver are notified.
- If a transfer fails during settlement (e.g., system error), the transaction is marked FAILED, the sender's available balance is refunded, and the sender is notified.
- Each payment request must carry a unique `request_id` generated by the client. If the same `request_id` is sent again (e.g., network retry), the system must return the original result instead of processing it again. This prevents double payments.

4. Fraud Controls

Every send-money request must pass through the following checks before processing:

Daily Transaction Limit A user cannot send more than ₹50,000 in a single day. If the transaction would exceed this limit, it must be blocked and a fraud alert must be created.

Duplicate Transaction Detection If a user tries to send the exact same amount to the exact same recipient within 2 minutes of a previous transaction, it must be blocked as a likely duplicate. A fraud alert must be created.

High-Value Transaction Flag If a single transaction exceeds ₹8,000, the transaction is allowed to proceed normally (enters PENDING settlement) but a fraud alert must be created for admin review. The admin has until the settlement date (3 business days) to review the alert and decide:

- **Block** — the transaction is marked FAILED, the sender's available balance is refunded, and both parties are notified.
- **No action** — the transaction settles normally on the settlement date. The admin missed their review window.

All fraud alerts must land in a queue visible to admins.

5. Transaction History

- Users must be able to view a mini-statement showing the last 10 transactions.
- Users must be able to view their full transaction history with pagination and filters for:
    - Date range
    - Type (sent / received / all)
    - Status (success / pending / failed / all)
- Pending transfers must clearly show:
    - Current status (Pending)
    - Expected settlement date

    - Days remaining until settlement
- Clicking on a transaction must show full details including ledger entries. For pending transfers, only the sender's DEBIT entry exists until settlement.
- Users must be able to see their personal stats: total sent, total received, and this month's transaction count.

6. Notifications

The system must generate in-app notifications for the following events:

Event	Who Gets Notified	Example
Transfer initiated	Sender	"₹2,000 transfer to Rahul initiated. Expected settlement: Oct 30"
Transfer settled	Sender	"₹2,000 transfer to Rahul has been settled"
Transfer settled	Receiver	"₹2,000 received from Priya"
Transfer failed	Sender	"₹2,000 transfer to Rahul failed. Amount refunded to your wallet"
Transaction blocked	Sender	"Transaction blocked: daily limit exceeded"
Fraud alert raised	Admin	"High-value transaction flagged: ₹9,000 by User #42"
Account locked	User	"Your account has been locked due to multiple failed login attempts"
Account unlocked	User	"Your account has been automatically unlocked"

- Users must see an unread count on a bell icon in the header.
- Users must be able to view all notifications and mark them as read.

7. Scheduled Background Jobs

The following jobs must run automatically on a schedule with no manual trigger:

ACH Settlement Processor Runs every hour. Finds all transactions where status is PENDING and `settlement_due_date` is on or before today. For each eligible transaction:

- Credits the receiver's available balance.
- Creates the CREDIT ledger entry for the receiver.
- Updates the transaction status from PENDING to SUCCESS.
- Sends settlement notifications to both sender and receiver.
- If any individual settlement fails, marks that transaction as FAILED, refunds the sender, and continues processing the rest.

Each settlement batch must be logged — how many transactions were settled, how many failed, total amount settled.

Daily Limit Reset Runs at midnight every day. Resets the daily spending counter for all users to ₹0 so they can transact again the next day.

Account Auto-Unlock Runs every 5 minutes. Finds all accounts that have been locked for more than 30 minutes and unlocks them. Sends an "account unlocked" notification to each affected user.

Daily Summary & Reconciliation Runs at 11 PM every day. Aggregates the day's activity into a summary:

- Total number of transfers initiated and their total amount
- Total number of transfers settled and their total amount
- Number of fraud alerts raised
- Number of new user registrations
- Total pending settlement amount across all users

Also performs a reconciliation check — for settled transactions, the sum of all DEBIT ledger entries must equal the sum of all CREDIT entries. Pending transactions are excluded from this check since the credit entry hasn't been created yet. If there is a mismatch among settled transactions, the system must create a critical notification for the admin.

Every job run must be logged with start time, end time, success/failure status, records processed, and error message if any.

8. Admin Panel

Users with the ADMIN role must have access to an admin section with the following:

Dashboard — Total registered users, today's transfers initiated count, today's transfers settled count, today's transaction volume (₹), total pending settlement amount, pending fraud alerts count, and a line chart showing daily transaction volume over the last 30 days.

User List — View all users with their wallet balance, wallet status, and registration date. Clicking on a user shows their recent transactions. Admin can freeze or unfreeze a user's wallet from here.

Transaction Monitoring — View all transactions across the system with filters (date, status, amount range). Clicking a transaction shows full details with ledger entries. Pending transactions must be clearly distinguishable.

Settlement Monitor — View all pending transactions with their expected settlement dates. Show a summary of how many are due today, due tomorrow, and overdue. Admin can manually trigger settlement for a specific transaction if needed (e.g., after resolving a system issue that caused a failure).

Fraud Review — View all pending fraud alerts. For high-value transaction alerts, admin can either "Block" the transaction (fails it and refunds the sender) or "Dismiss" the alert. Alerts for transactions that have already settled are read-only.

Job Monitor — View the last 20 scheduled job runs with their status and stats. The settlement job should show settled count, failed count, and total amount per run.

Reconciliation — Display today's total debits vs total credits for settled transactions. Show pending settlement amount separately. Flag any mismatches.

All admin actions must be written to an audit log.

Business Rules

Rule	Value
Daily transaction limit per user	₹50,000
Duplicate detection window	2 minutes
High-value flag threshold	₹8,000
Settlement period	3 days
Account lock after failed logins	5 attempts
Auto-unlock after	30 minutes
Initial wallet balance	₹0
Transaction PIN	4 digits
Transaction statuses	PENDING → SUCCESS / FAILED / DECLINED
Roles	USER, ADMIN

Database

11 tables: `users`, `login_attempts`, `wallets`, `transactions`, `ledger_entries`, `fraud_alerts`, `notifications`, `audit_log`, `scheduled_job_log`, `daily_summary`, `settlement_batches`

- `settlement_batches` — tracks each settlement job run with counts and amounts